

Module description: Data Analytics		
Module Code	w.BA.XX.3DA-WIN.XX	
ECTS Credits	6	
Language of Instruction/Examination	English	
Module Description	This module provides students with the foundations and basic concepts of data analytics. Students learn to create e2e data analytics workflows and perform their own practice-oriented data analytics projects. They become familiar with the necessary tools and programming languages for data analytics. The module builds on the content of the "Statistics" module and provides the basis for the modules "AI Applications" and "Business Intelligence."	
Organizational Unit	Institut für Wirtschaftsinformatik	
Module Coordinator	Mario Gellrich	
Deputy Module Coordinator	Alexander Walter	
Legal Framework	Academic Regulations BSc dated 29.01.2009, for the degree programs in Business Administration, International Management, Business Information Technology, Business Law, Business Law and Applied Law, first adopted on 12.05.2009	
Module Category	Module Type Compulsory	Program Phase Main Study Period
Prerequisite Knowledge	• Basic programming experience using the Python programming language • Previous knowledge of statistics	
Contribution to Learning Objectives (by the concerned Module)	• Professional Competence • Methodological Competence • Social Competence • Self-Competence	
Contribution to Learning Objectives	Professional Competence • Knowing and Understanding Content of Theoretical and Practical Relevance • Apply, Analyze, and Synthesize Content of Theoretical and Practical Relevance • Evaluate Content of Theoretical and Practical Relevance Methodological Competence • Problem-Solving & Critical Thinking • Scientific Methodology • Work Methods, Techniques, and Procedures • Information Literacy • Creativity & Innovation Social Competence • Written Communication • Oral Communication • Teamwork & Conflict Management • Intercultural Insight & Ability to Change Perspective Self-Competence • Self-Management & Self-Reflection • Ethical & Social Responsibility • Learning & Change	

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Module Learning Objectives	Students... • can recognize the potential and benefits of data in a business context. • know common tools and programming languages for data analytics and can apply them in practice-oriented projects. • can create e2e data analytics workflows for practice-oriented projects. • know common data sources and can apply methods for data collection including web scraping and web APIs. • know and can work with different data types, data structures, and file formats in Python. • can import and export data in different file formats and read, query, and write SQL databases from Python. • can clean, organize, enrich, and create new data and simulate data in Python. • can analyze data using non-graphical and graphical exploratory data analysis (EDA) methods. • can graphically represent and analyze geographical data. • can perform univariate data analysis using frequency tables and measures of central tendency and variability. • can perform bivariate data analysis using cross tables, correlation analysis, and statistical tests for the comparison of central tendencies. • can perform multivariate data analysis using regression models (OLS, RT), classification models (CT, RF), and clustering (k-means). • can perform sentiment analysis based on social media and other textual data. • can prepare, present, and explain data analysis results in a professional and comprehensible way.					
Module Content	• Introduction to data analytics: Basic concepts, methods, and applications • Data sources and data collection methods: OpenData, web scraping, and web APIs • Data types and structures in Python and common file formats • Import and export of data in Python and working with SQL databases • Clean, organize, enrich, and create new data and simulate data in Python • Graphically and non-graphically exploratory data analysis (EDA) • Graphical representation and analysis of geographical data • Statistical hypothesis testing (I): Paired & unpaired sample tests, analysis of contingency tables • Statistical hypothesis testing (II): Correlation analysis, analysis of variance (ANOVA) • Multivariate data analysis and predictive modeling: regression (linear regression, regression trees) • Multivariate data analysis and predictive modeling: classification (classification trees, random forests) • Multivariate data analysis and predictive modeling: clustering (k-means) • Text mining and sentiment analysis • Presenting and sharing data analysis results					
Links to other modules	<table><tr><th>Linked Module</th></tr><tr><td>w.BA.XX.3BI-WIN.XX / Business Intelligence</td></tr><tr><td>w.BA.XX.3KIA-WIN.XX / AI Applications</td></tr><tr><td>w.BA.XX.3Stat-WIN.XX / Statistics</td></tr></table>		Linked Module	w.BA.XX.3BI-WIN.XX / Business Intelligence	w.BA.XX.3KIA-WIN.XX / AI Applications	w.BA.XX.3Stat-WIN.XX / Statistics
Linked Module						
w.BA.XX.3BI-WIN.XX / Business Intelligence						
w.BA.XX.3KIA-WIN.XX / AI Applications						
w.BA.XX.3Stat-WIN.XX / Statistics						
Digital Learning Resources	• Teaching Materials • Practice and Application Exercises (with Key)					
Methods of Instruction	• Lecture • Project Work • Application Tasks • Exercises • Literature Review	Social Settings Used: • Individual Work • Group Work				

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Type of Instruction		Classroom Instruction	Guided Self-Study	Autonomous Self-Study
	Large Class	28 h	-	
	Small Class	28 h	56 h	
	Group Instruction	-	-	
	Practical Work	-	-	
	Seminar	-	-	
	Total	56 h	56 h	68 h

Performance Assessment	End-of-module exam	Form	Length (min.)	Weighting
	Written exam	open book	60	60.00%
	Permitted Resources	Spec. calculator acc. to leaflet "Utilities"	With dictionary	

	Assessment	Format	Length (min.)	Weighting
Project work <i>The grade for the project work is made up of two parts: 1.) Technical component: This includes the entire data analytics process, including the programming work and required submissions on Moodle. 2.) Live presentation in class: The group presents their project concept, data, and methods orally in front of the class. This does not cover the finished project work or final results. Note: The scheduled time of 30 min. is a rough estimate, covering the final presentation of the project in the form of a video (submitted on Moodle) plus the oral presentation in class. The actual time required depends on group size. Further details will be communicated on Moodle at the beginning of the semester.</i>	Grade	Group work	30	40.00%

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Classroom Attendance Requirement	Other Attendance requirements apply to the specific days scheduled for the individual project presentations in class. On these days, the members of the presenting groups are required to attend on-site for their oral presentation. A schedule with the exact dates will be communicated in good time in class and via the Moodle learning platform.
Compulsory Reading	• The literature issued in class or made available on the Moodle teaching platform is compulsory reading.
Recommended Reading	• -
Comments	Supplementary examinations may not be set, and assessed work may not be resubmitted.